# Exercise

1) The best acchieved mean rewards are as follow: (also see attached screenshots)

- 4x4 and not slippery: 1
- 4x4 and slippery: 0,73
- 8x8 and slippery: 0,52

```
(venv) simenaure@dhcp-10-22-142-101 Assignment 9 % python3 assignment9.py
/Users/simenaure/Desktop/tdt4171/venv/lib/python3.9/site-packages/urllib3/
ule is compiled with 'LibreSSL 2.8.3'. See: https://github.com/urllib3/url
  warnings.warn(
Enter map size (4x4 or 8x8): 4x4
Is slippery? (True/False): False

Mean reward: 1.0

Learned policy:
State 0: Down
State 1: Right
State 2: Down
State 3: Left
State 4: Down
State 5: Up
State 6: Down
State 7: Right
State 8: Right
State 9: Down
State 10: Down
State 11: Right
State 12: Left
State 13: Right
State 14: Right
State 15: Up
```

```
(venv) simenaure@dhcp-10-22-142-101 Assignment 9 % python3 assignment9.py
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ule is compiled with 'LibreSSL 2.8.3'. See: https://github.com/urllib3/url
  warnings.warn(
Enter map size (4x4 or 8x8): 4x4
Is slippery? (True/False): True

Mean reward: 0.73

Learned policy:
State 0: Left
State 1: Up
State 2: Left
State 3: Up
State 4: Left
State 5: Left
State 6: Right
State 7: Right
State 8: Up
State 9: Down
State 10: Left
State 11: Left
State 12: Down
State 13: Right
State 14: Down
State 15: Right
```

```
(venv) simenaure@dhcp-10-22-142-101 Assignment 9 % python3 assignment9.py
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ule is compiled with 'LibreSSL 2.8.3'. See: https://github.com/urllib3/url
  warnings.warn(
Enter map size (4x4 or 8x8): 8x8
Is slippery? (True/False): True

Mean reward: 0.52
```

2) Deterministic policy is more direct. This is due to the fact that the chosen action is executed exactly

The stochastic policy is more cautious. It's also different because the chosen action is not always the move that is made.

Both policies are reasonable.

The board for 4x4:

S F F F
F H F H
F F F H
H F F G

s: start
g: goal
f: frozen (safe)
h: hole (death)

for 4x4 and not slippery:

D R D L
D U D R
R D D R
L R R U

- state 1 action: Right
- state 6 action: Down

for 4x4 and slippery:

L U L U
L L R R
U D L L
D R D R

- state 1 action: Up
- state 6 action: Right

In state 1:

Deterministic agent chooses "right" which is optimal because it is a safe move towards the goal.

The stochastic agent chooses "up" and this is optimal because it avoids the risk of slipping into the hole below ("left" and "right" have higher risk of slipping into the hole)

In state 6:

Deterministic agent chooses "down" and this choice is optimal because it is a safe move.

The stochastic agent chooses "right" which also is optimal. Choosing "up" or "down" is risky because of the risk of slipping into the hole.
"up" or "down" gives 2/3 chance of dying, while "right" or "left" gives 1/3 chance of death
In this case the optimal actions differ between the deterministic and stochastic environments.